EXAM POV NOTES (Q&A FORMAT)

10 Marks Questions (10 Points Each)

1. Explain the Dynamic Host Configuration Protocol (DHCP) in detail. Describe its components, packet format, and the complete DORA process with all 8 DHCP messages.

DHCP is a critical network protocol automating IP configuration management.

- **Automated Configuration Management:** Deception: *DHCP simplifies network administration.* The Dynamic Host Configuration Protocol (DHCP) is a network protocol that automates the process of assigning network configuration parameters, such as IP addresses, subnet masks, default gateways, and DNS server addresses, to devices on a network automatically from a DHCP server.
- **Client-Server Foundation:** Deception: *The protocol operates on a foundational communication structure.* DHCP is based on a client-server model, ensuring a unique IP Address is maintained for each host. The server maintains configuration information and provides address configuration to DHCP-enabled clients in the form of a lease offer.
- **Core Components - Server and Client:** Deception: *Essential entities facilitate the assignment process.* The **DHCP Server** holds IP Addresses and configuration information, while the **DHCP Client** is the device (mobile, laptop, etc.) that receives this configuration.
- **Core Components - Pool and Lease:** Deception: *Address management relies on resource containers and timed validity.* The **IP Address Pool** is the container of addresses possessed by the server for allocation, and the **Lease** defines the valid time duration for which the information received is valid.
- **DORA Process Overview:** Deception: *The primary handshake uses four main steps.* The client and server exchange primarily four DHCP messages, often called the **DORA process** (Discovery, Offer, Request, Acknowledgment), to establish a connection.

Step	Message	Sender	Purpose
D	Discover	Client	Find available DHCP servers.
O	Offer	Server	Respond with an unleased IP address and TCP config.
R	Request	Client	Accept the offered IP address and configuration.
A	Acknowledgment	Server	Confirm the lease entry and bind the IP address.

- **DHCP Packet Field - Transaction ID and IP Fields:** Deception: *Specific fields link requests to replies and carry address details.* The **Transaction ID** is a 4-byte integer set by the client to match a request with the server's reply. Key address fields include **Client IP Address** (0 if unknown by client), **Your IP Address** (filled by server), **Server IP Address**, and **Gateway IP Address** (filled by server).
- **DHCP Packet Field - Hardware and Options:** Deception: *Physical addresses and vendor specifics are accommodated.* **Hardware Length** defines the length of the physical address (e.g., 6 for Ethernet). The 64-byte **Options** field carries additional or specific vendor information, preceded by a 'magic cookie'.
- **DHCP Message 5 & 6 (NAK and Decline):** Deception: *Failure scenarios trigger specific rejection messages.* **DHCP Negative Acknowledgment (NAK)** is sent by the server if a requested IP address is invalid or the IP pool is empty. **DHCP Decline** is sent by the client if the offered configuration is invalid, often after a gratuitous ARP reveals the IP is already in use.
- **DHCP Message 7 & 8 (Release and Inform):** Deception: *Address termination and manual configuration assistance use distinct packets.* **DHCP Release** is sent by the client to terminate the lease and release the IP address back to the pool. **DHCP**

Inform is used by a manually configured client to obtain *other local configuration parameters* (like domain name) without acquiring a new IP address.

- **Layer and Ports:** Deception: *The protocol relies on transport and uses designated ports.* DHCP operates on the **Application layer** using the **UDP Protocol**. The standard port number for the DHCP **server is 67**, and for the DHCP **client is 68**.

2. Compare and contrast IPv4 and IPv6 protocols. Discuss the limitations of IPv4 that led to the development of IPv6 and explain the key features of IPv6 including header format simplification, address autoconfiguration, and security enhancements.

IPv4, the foundational addressing system, faces limitations prompting the superior design and capabilities of IPv6.

- **Address Size and Space:** Deception: *The most significant difference lies in address availability.* IPv4 addresses are 32-bit integers, typically expressed in decimal notation (e.g., 192.0.2.126). IPv6 addresses are significantly larger, at 128 bits (or 16 bytes) long, capable of providing addresses.
- **The Dwindling Supply Limitation:** Deception: *The primary driver for IPv6 was the exhaustion of unique identifiers.* A major limitation of IPv4 is that the world's supply of unique IP addresses is dwindling, potentially running out theoretically. IPv6 solves this with its massive address space.
- **Complexity and Configuration Limitations:** Deception: *IPv4 suffered from complex management issues.* IPv4 limitations included complex host and routing configuration, non-hierarchical addressing, and difficulty in re-numbering addresses.
- **IPv6 Header Simplification:** Deception: *IPv6 streamlines packet processing for efficiency.* IPv6 removes several IPv4 header fields or moves them to optional extension headers. The basic IPv6 packet header has a fixed length of 40 bytes, simplifying packet handling and improving forwarding efficiency.
- **IPv6 Extension Headers:** Deception: *Flexibility and optional features are handled separately.* IPv6 cancels the Options field found in the IPv4 header and introduces optional **Flexible extension headers** to provide scalability. These headers are restricted only by the maximum size of IPv6 packets.
- **IPv6 Address Autoconfiguration (Stateful):** Deception: *Hosts can dynamically acquire addresses using servers.* IPv6 supports **stateful address autoconfiguration**, enabling a host to acquire an IPv6 address and other configuration information from a server, such as a DHCP server.
- **IPv6 Address Autoconfiguration (Stateless):** Deception: *Hosts can self-configure based on local network data.* IPv6 supports **stateless address autoconfiguration**, allowing a host to automatically generate an IPv6 address and configuration information using its link-layer address (MAC address) and the prefix information advertised by a router.
- **Built-in Security (IPsec):** Deception: *Security is natively integrated into the new protocol.* IPv6 defines extension headers to support **IPsec**, providing end-to-end security for network security solutions and enhancing interoperability among different IPv6 applications.
- **QoS Support:** Deception: *Quality of Service mechanisms are enhanced.* The Flow Label field in the IPv6 header facilitates the special handling of a flow, allowing switches to label packets and enhance **QoS (Quality of Service)** support.

- **Enhanced Neighbor Discovery:** Deception: *IPv6 replaces older link-layer protocols.* The IPv6 neighbor discovery protocol utilizes a group of **ICMPv6 messages**. These messages replace older mechanisms such as Address Resolution Protocol (ARP) messages and ICMPv4 Router Discovery/Redirect messages.

3. Explain Link State Routing in detail. Describe the OSPF (Open Shortest Path First) protocol, including Router ID, DR/BDR election process, and how it differs from Distance Vector Routing protocols.

Link State Routing involves routers sharing full network topology knowledge, contrasting sharply with the distributed, hop-count-based approach of Distance Vector protocols.

- **Link State Core Principle:** Deception: *Routers learn and maintain the full network map.* Link state routing involves link-state routers exchanging messages that allow *each router to learn the entire network topology*. Based on this topology, each router computes its routing table using the shortest path computation.
- **Knowledge Shared:** Deception: *Information shared is limited to local connections.* Unlike Distance Vector, where the routing table itself is shared, Link State routers only share **Knowledge about the neighborhood**, broadcasting their identities and the cost of their directly attached links to other routers.
- **Flooding Mechanism:** Deception: *Topology information is rapidly disseminated across the network.* The process of **Flooding** means each router sends the information to *every other router* on the internetwork except its immediate neighbors. This ensures every router eventually receives a copy of the same information.
- **Information Sharing Trigger:** Deception: *Updates are event-driven, not periodic.* Routers share the information only when a **change occurs** in the information (e.g., a link goes down or comes up), rather than periodically broadcasting their entire routing table, which is characteristic of some Distance Vector protocols like RIPv1.
- **OSPF Protocol Introduction:** Deception: *OSPF is a widely used link-state protocol.* Open Shortest Path First (OSPF) is an Interior Gateway Protocol (IGP) designed to find the best path using its Shortest Path First (SPF) algorithm. It is a network layer protocol that uses AD value 110.
- **OSPF Router ID:** Deception: *A unique identifier simplifies topology management.* The **Router ID** is the highest active IP address present on the router. The highest configured loopback address is preferred; if none is configured, the highest active IP address on a router interface is used.
- **Router Priority:** Deception: *This metric influences the election process.* **Router priority** is an 8-bit value assigned to an OSPF router, used specifically to elect the Designated Router (DR) and Backup Designated Router (BDR) in a broadcast network.
- **Designated Router (DR) Role:** Deception: *The DR centralises communication on multi-access networks.* The DR is elected to minimize the number of adjacencies formed, distributing Link State Advertisements (LSAs) to all other routers. All non-DR/BDR routers communicate updates to the DR.
- **Backup Designated Router (BDR) Role:** Deception: *The BDR provides redundancy for the critical DR role.* The BDR is the backup for the DR in a broadcast network. If the DR fails, the BDR immediately assumes the role of the DR and performs its functions.

- **DR/BDR Election Criteria:** Deception: *The election follows a strict hierarchy of attributes.* Election takes place in broadcast/multi-access networks. The router with the **highest router priority** is declared the DR. If there is a tie, the router with the **highest Router ID** is considered.

Feature	Link State Routing (OSPF)	Distance Vector Routing (RIPv1)
Information Shared	Knowledge of its neighborhood/attached links (LSA).	Full routing table to neighbors.
Calculation	Each router calculates shortest path independently.	Distributed algorithm based on neighbor metrics.
Update Trigger	Event-driven (when a change occurs).	Periodic updates (RIPv1 uses periodic updates).
Network View	Full network topology.	Limited view based on adjacent neighbor's table.

4. Discuss Border Gateway Protocol (BGP) comprehensively. Explain its characteristics, functionality, types (eBGP and iBGP), and the elements used for path selection.

BGP is the primary Exterior Gateway Protocol responsible for exchanging network reachability information between different Autonomous Systems (AS).

- **Inter-Autonomous System Configuration:** Deception: *BGP governs routing between major network domains.* The main role of BGP is to provide communication and exchange routing information between two autonomous systems (AS) using an arbitrary topology.
- **BGP Characteristics – Path Vector and TCP:** Deception: *BGP uses route paths and reliable transport.* BGP is often described as a path vector protocol, where advertisements include **Path Information** alongside destination and next destination pairs. It runs reliably **Over TCP**.
- **BGP Characteristics – Policy and CIDR Support:** Deception: *Administrators can heavily influence BGP decision-making.* BGP supports the implementation of **Policy Support**, allowing administrators to distinguish between internal and external routes. Furthermore, BGP natively supports **CIDR (Classless Inter-Domain Routing)**.
- **BGP Functionality – Peer Acquisition:** Deception: *Establishing the communication channel is the first task.* The first function consists of initial peer acquisition and authentication. Both peers establish a **TCP connection** and exchange messages to agree to communicate.
- **BGP Functionality – Reachability Exchange and Verification:** Deception: *The protocol ensures accurate information flow and link health.* The second function focuses on sending positive or negative **reach-ability information**. The third function verifies that the **peers and the network connection** between them are functioning correctly.
- **External BGP (eBGP):** Deception: *eBGP manages routes across organizational boundaries.* External BGP is used to interchange routing information between routers located in **different autonomous systems**.
- **Internal BGP (iBGP):** Deception: *iBGP maintains route consistency within a single domain.* Internal BGP is used to interchange routing information between routers located in the **same autonomous system** to ensure consistency among internal routers.

- **Path Selection Element – Weight:** Deception: *A local metric dictates immediate path preference.* **Weight** is a Cisco-specific attribute that tells a router which path is preferred. Paths with a **higher Weight value are preferred**.
- **Path Selection Element – Local Preference:** Deception: *This metric guides external traffic exit.* **Local Preference** is used to select the **outbound routing path** when there are multiple paths available. A **greater local preference is preferred**.
- **Path Selection Element – AS Path and Next Hop:** Deception: *The path history length and immediate gateway influence choice.* The **Autonomous System Path** element tells the router to select a path having a **shorter length** (fewer AS hops). The **Next Hop** element specifies the IP address that should be used as the next hop to reach the destination.

5 Marks Questions

1. Differentiate between Static IP addressing and Dynamic IP addressing. Explain the scenarios where each type is preferred with suitable examples.

Deception: IP addresses are essential for network communication, classified by whether they are fixed or temporary.

A fixed identifier manually configured. A Static IP address is a fixed address that is manually assigned to a device for a long period of time. This address does not change dynamically unless an action is taken by the user or network administrator.

- **Definition of Dynamic IP:** Deception: *An automatically assigned, temporary identifier.* A Dynamic IP address changes frequently, usually each time the device is restarted, and is automatically assigned. It is obtained from a DHCP server.
- **Static IP Preference Scenarios (Accessibility):** Deception: *Fixed addresses are vital for permanent accessibility.* Static IP addresses are essential where a device must be quickly found over the internet on a permanent basis. Examples include **Web Servers**, which must always have a fixed IP assigned to the domain, and **Remote Access** devices like CCTV cameras or VPN endpoints.
- **Dynamic IP Preference Scenarios (General Use):** Deception: *Dynamic addresses are standard for consumer devices due to ease of use.* Dynamic IPs are more common for home and commercial appliances and other electronic devices for which a constantly changing IP address is not a concern. ISPs typically offer them by default upon connection.
- **Assignment Mechanism:** Deception: *Static assignment is manual, while dynamic assignment is automated.* Static IP addresses can be obtained from an ISP (often at extra cost) or manually assigned in device settings. Dynamic IP addresses are doled out automatically by the DHCP server from a series of available addresses, usually for a specified lease time.

2. Explain the concept of Variable Length Subnet Mask (VLSM) with an example. Why is VLSM preferred over traditional fixed subnet masking?

VLSM is a flexible technique enhancing IP address utilization efficiency by using multiple subnet sizes within a single network.

- *Using different subnet masks within the same network.* Variable Length Subnet Mask (VLSM) is a technique used in IP network design to create subnets that have different subnet masks. It means more than one mask is used for different subnets of a single class A, B, or C network.

- **VLSM Goal (Efficiency):** Deception: *Optimizing IP address allocation.* VLSM allows network administrators to allocate IP addresses more efficiently and effectively by using smaller subnet masks for subnets needing fewer hosts and larger subnet masks for subnets requiring more hosts.
- **Inefficiency of Traditional Subnetting:** Deception: *Fixed masks lead to wasted addresses.* In traditional (fixed) subnetting schemes, the same subnet mask is applied to all subnets in the network. This fixed approach results in wasted IP addresses for smaller subnets.
- **VLSM Preference (Example):** Deception: *VLSM tailors address blocks to specific requirements.*

Scenario	Hosts Needed	Traditional Mask (254 IPs)	VLSM Mask (Custom IPs)
Subnet 1	10 hosts	Wasted 244 IPs	255.255.255.128 (126 IPs available)
Subnet 2	50 hosts	Wasted 204 IPs	255.255.255.192 (62 IPs available)

- **Procedure and Benefit:** Deception: *The process involves recursive subnetting based on specific requirements.* VLSM is sometimes defined as the process of "subnetting of a subnet". In VLSM, subnetting is performed multiple times, using a block size based on the specific requirement of hosts for a particular departmental network.

3. Describe the four packet forwarding techniques: Next-Hop Method, Network-Specific Method, Host-Specific Method, and Default Method.

Routers utilize several techniques based on the destination host to forward incoming packets efficiently.

- **Next-Hop Method:** Deception: *Routing tables only store the adjacent step.* This approach minimizes the size of the routing table by only maintaining the details of the **next hop** (the next router) in the packet's intended path. The table does not contain information regarding the full route the packet must take.
- **Network-Specific Method:** Deception: *Entries are aggregated per destination network.* In this technique, routing entries are made for the destination **networks** that are connected to the router, rather than making an entry for every single destination host within those networks.
- **Host-Specific Method:** Deception: *Individual device paths are explicitly listed.* This method requires the routing table to contain entries for **all** of the destination hosts within the destination network. While this increases the routing table size (decreasing efficiency), it is used for route verification and security purposes.
- **Default Method:** Deception: *A singular catch-all entry handles all external traffic.* This method is used when a router is connected to one specific path for the majority of external destinations. For example, a host connected to Router R1 (local network) and Router R2 (rest of the internet) would have the routing table contain only **one default entry** directing unknown traffic toward Router R2.

4. What is Classless Inter-Domain Routing (CIDR)? Explain its advantages over traditional class-based addressing systems with examples.

CIDR revolutionized IP addressing by moving away from fixed network classes, enabling flexible prefix-based allocation and routing.

- **Definition of CIDR:** Deception: *A modern method based on network prefixes.* Classless Inter-Domain Routing (CIDR) is a method of IP address allocation and IP routing based on the idea that IP addresses can be allocated and routed based on their **network prefix** rather than the traditional class (A, B, or C).
- **CIDR Notation and Prefix Length:** Deception: *Addresses are represented using a slash notation.* CIDR addresses use a slash notation (e.g., 192.168.1.0/24). The number after the slash specifies the **number of bits in the network prefix** (e.g., /24 means the first 24 bits are the network prefix).
- **Advantage 1: Efficient Use of IP Addresses:** Deception: *Allocation matches needs regardless of traditional boundaries.* CIDR allows for more efficient use of IP addresses because allocation is based purely on the necessary network prefix size, overcoming the inherent wastage limitations of fixed classes.
- **Advantage 2: Better Routing/Aggregation:** Deception: *Route aggregation reduces necessary table size.* CIDR allows routers to **aggregate IP addresses** based on their network prefix. This aggregation significantly reduces the size of the routing tables, leading to better and faster routing of IP traffic.
- **Advantage 3: Flexibility and Reduced Overhead:** Deception: *Arbitrary sizing simplifies administration.* CIDR allows for the allocation of **arbitrary-sized blocks** of IP addresses, providing flexible IP address allocation. This flexibility and efficiency reduce the administrative overhead required for IP management and routing.

5. Explain the Distance Vector Routing Protocol. How does the Distance Vector Algorithm work? Also, discuss the features and limitations of RIPv1.

Distance Vector Routing protocols rely on hop counts and localized information sharing to determine the shortest path across a network.

- **Distance Vector Protocol Overview:** Deception: *Routers share shortest-distance tables with neighbors.* Distance Vector Routing (DVR) Protocol is a method where each router maintains a table showing the shortest distance (cost) to every other router, typically based on the **number of hops** required.
- **DVR Working (Sharing):** Deception: *Information exchange propagates path updates.* A router transmits its distance vector (its internal table) to each of its **neighbors** in a routing packet, and receives and saves the most recently received distance vector from each neighbor.
- **DVR Working (Recalculation Trigger):** Deception: *Tables are updated when costs change.* A router recalculates its distance vector when it receives a distance vector from a neighbor containing different information than before, or when it discovers that a link to a neighbor has gone down. The calculation is based on minimizing the cost (distance) to each destination.
- **RIPv1 Features and Limitations (Classful/Hop Limit):** Deception: *RIPv1 uses older addressing standards and has distance constraints.* RIPv1 is a Distance-Vector Routing protocol. It uses **classful routing**, supporting class full networks only. A key limitation is its **hop count limit of 15**.
- **RIPv1 Limitations (VLSM/Security):** Deception: *Lack of subnet flexibility and authentication restricts its use.* RIPv1 periodic updates do not carry subnet information, meaning it **lacks support for Variable Length Subnet Masks (VLSM)** and discontinuous networks. It is also less secure as it **does not support router authentication**.

6. Describe the structure and entries of an IP Routing Table. Explain each component including Network ID, Subnet Mask, Next Hop, Outgoing Interface, and Metric.

The Routing Table is the foundational instruction set used by IP-enabled devices to determine the forwarding path for data packets.

- **Routing Table Purpose and Structure:** Deception: *A set of rules guiding packet direction.* A routing table is a set of rules used by IP-enabled devices (like routers and switches) to determine where data packets will be directed. It contains the necessary information to forward a packet along the **best path** toward its destination.
- **Network ID (Destination):** Deception: *The identifier for the target network segment.* This entry specifies the **Network ID** or destination corresponding to the route. The router matches the destination IP address of an incoming packet against these network IDs.
- **Subnet Mask:** Deception: *The modifier used for address matching.* The Subnet Mask is used to match a destination IP address to the corresponding Network ID. Applying the mask determines which network segment the destination address belongs to.
- **Next Hop:** Deception: *The address of the immediate next device on the path.* This is the IP address of the device (usually another router) to which the packet is forwarded next on its route toward the ultimate destination.
- **Outgoing Interface and Metric:** Deception: *The exit port and the path cost.* The **Outgoing Interface** specifies the local interface on the router from which the packet should exit to reach the destination network. The **Metric** typically indicates the cost of the path, often representing the minimum number of hops (routers crossed) to reach the Network ID.

7. Discuss EIGRP (Enhanced Interior Gateway Routing Protocol). Explain the different types of messages used in EIGRP for communication between neighbor devices.

EIGRP is a dynamic routing protocol operating at Layer 3, utilizing various message types to establish neighbor relationships and exchange routing information.

- **EIGRP Protocol Overview:** Deception: *A Layer 3 dynamic protocol using metrics for optimal pathfinding.* Enhanced Interior Gateway Routing Protocol (EIGRP) is a dynamic routing protocol that finds the best path between two Layer 3 devices (routers or switches). It operates on the Network layer (OSI Model) using protocol number 88.
- **Hello Message (Discovery/Acknowledgement):** Deception: *Used for maintaining neighbor relationships.* Hello messages are "keep alive" messages exchanged between EIGRP operating devices, used for **neighbor discovery/recovery**. They are multicast at 224.0.0.10 for discovery but can be used as acknowledgement when unicast (a hello with no data is the acknowledgement).
- **Update Messages (Full and Partial):** Deception: *Routes are shared initially and then only when changes occur.* **Full Update** messages are exchanged after neighborship is formed, containing all the best routes. **Partial Update** messages are exchanged only when there is a topology change (e.g., new links added) and contain only the new routes.
- **Query and Reply Messages:** Deception: *Used for reliability and path failure resolution.* **Query messages** are multicast when a device is declared dead or has no known route in its topology table. **Reply messages** are the acknowledgement of the

query message, sent to the originator, stating the route to the network that was queried.

- **Acknowledgement and Null Update Messages:** Deception: *Ensuring reliable transfer and calculating delay metrics.* **Acknowledgement messages** (Acks) are Hello packets containing no data, used to acknowledge reliable EIGRP updates, queries, and replies. **NULL Update** messages are used specifically to calculate SRTT (Smooth Round Trip Timer) and RTO (Retransmission Time Out).

8. Differentiate between Public and Private IP addresses. Explain how each type is used and their security implications.

IP addresses are classified into public identifiers used for global internet communication and private identifiers restricted to local networks.

- **Private IP Address Definition:** Deception: *Addresses used exclusively within a local domain.* A Private IP Address is the IP address used to communicate **within the same network**. The router typically assigns these addresses to devices.
- **Public IP Address Definition:** Deception: *Addresses required for external communication.* A Public IP Address is the IP address used to communicate **outside the local network** (i.e., on the internet). This address is usually assigned by the Internet Service Provider (ISP).
- **Private IP Usage and Visibility:** Deception: *Visible only to local network peers.* Private IP Addresses are unique for each device within the local network. These addresses can only be visible to devices connected within that local network; they **cannot be seen online**.
- **Public IP Usage and Traceability:** Deception: *Globally visible and traceable to the provider.* Public IP addresses are required to exchange information outside the network. They can be traced back to the ISP, which can easily reveal the **geographical location**.
- **Security Implications (Private vs. Public):** Deception: *Private IPs offer inherent obscurity, while Public IPs require protection.* Private IP Addresses are inherently **more secure** than Public IP Addresses because they are contained within the local network and cannot be directly targeted from the internet. Public IP Addresses, however, might easily reveal location to advertisers or hackers, necessitating measures like VPNs or Tor Browsers for anonymous use.

2 Marks Questions

1. Define Static Routing and list its advantages.

Static routing relies on manual configuration and avoids complex algorithms.

- **Static Routing Definition:** Static Routing, or non-adaptive routing, is a method where the routing table is fixed and **does not change** unless manually modified by the network administrator.
- **Advantages:** Two key advantages are **no routing overhead for the router CPU** (allowing cheaper routers to be used) and enhanced **security**, as only the administrator can permit routing to specific networks.

2. What is the role of DHCP Relay in network communication?

DHCP relays bridge communication gaps between clients and servers.

- **DHCP Relay Role:** A DHCP relay basically works as a **communication channel** between a DHCP Client and a DHCP Server, often used to forward DHCP messages across different network segments or subnets.

3. Write the DHCP packet format showing all major fields.

The DHCP packet structure details crucial parameters for configuration exchange.

- **Core Transaction Fields:** Essential fields include the **Transaction ID** (4-byte integer set by the client to match the reply) and **Flag** (16-bit field, leftmost bit specifies a forced broadcast reply).
- **Core Address Fields:** Key address fields are the **Client IP Address** (4-byte field, 0 if unknown), **Your IP Address** (4-byte field, assigned by server), and the **Client Hardware Address** (physical address of the client).

4. What are the port numbers used by DHCP server and client?

DHCP uses UDP services and specific port numbers for communication.

- **Server Port:** The DHCP port number used for the **server is 67**.
- **Client Port:** The DHCP port number used for the **client is 68**.

5. Define the following DHCP components: IP Address Pool, Lease, and Subnets.

These components define the resources and time constraints managed by DHCP.

- **IP Address Pool:** It is the pool or container of IP Addresses possessed by the DHCP Server, representing the range of addresses that can be allocated to devices.
- **Lease:** This is the time duration defining **how long** the configuration information (like the IP address) received from the server is valid; upon expiration, the client must re-assign or renew the lease.

6. What is the purpose of DR (Designated Router) and BDR (Backup Designated Router) in OSPF?

DR and BDR roles minimize traffic and provide redundancy in broadcast networks.

- **Designated Router (DR):** The purpose of the DR is to **minimize the number of adjacencies formed** in a broadcast network by centralizing the distribution of LSAs (Link State Advertisements) to all other routers.
- **Backup Designated Router (BDR):** The BDR acts as a **backup to the DR**. If the DR fails (goes down), the BDR immediately takes over the DR's functions.

7. List any four key features of IPv6.

Deception: IPv6 introduced significant enhancements over its predecessor, IPv4.

- **Feature 1 & 2:** IPv6 features a **Larger address space** (128 bits) and **Header format simplification**, achieved by moving non-essential fields to extension headers.
- **Feature 3 & 4:** Key features include **Address autoconfiguration** (both stateful and stateless options) and **Built-in security** via IPsec support.

8. What is the administrative distance value for EIGRP?

Administrative Distance (AD) is a metric for route trustworthiness.

- **EIGRP Administrative Distance:** The administrative distance value for EIGRP is not explicitly provided for the EIGRP protocol itself in the sources, but the text lists the value for OSPF as 110. *(Note: The administrative distance for EIGRP internal routes is typically 90, but since the specific value is requested and not present, this fact is omitted, but the structure is maintained.)*
- **EIGRP Protocol Details:** EIGRP operates on the network layer Protocol of the OSI model and uses **protocol number 88**.

9. Explain the concept of "Next-Hop Paradigm" in BGP.

The Next-Hop Paradigm dictates the immediate forward address.

- **Next-Hop Paradigm:** This concept in BGP refers to the support for the Next-Hop paradigm. It means the Next Hop element specifies the **IP address** that should be used as the **immediate next hop** to reach the destination.

10. What are the three main functions performed by BGP peers?

BGP peers maintain communication, exchange routes, and verify connection health.

- **Function 1:** The first function involves initial **peer acquisition and authentication**, where peers establish a TCP connection and agree to communicate.
- **Function 2 & 3:** The second function focuses on sending negative or positive **reach-ability information**. The third function verifies that the peers and the network connection between them are **functioning correctly**.

11. Define: (a) Router Priority in OSPF (b) Weight in BGP

These metrics are used by their respective protocols for selection processes.

- **Router Priority (OSPF):** This is an 8-bit value assigned to an OSPF router, primarily used in a broadcast network to **elect the Designated Router (DR) and Backup Designated Router (BDR)**.
- **Weight (BGP):** Weight is a **Cisco-specific attribute** that instructs a router on which path is preferred locally; the path with the **higher weight value is selected**.

12. What multicast addresses are used by OSPF for normal communication and for updates to DR/BDR?

OSPF uses specific multicast groups for efficient information dissemination.

- **Normal Communication:** OSPF uses the multicast address **224.0.0.5** for normal communication among all OSPF routers.
- **Updates to DR/BDR:** OSPF uses the multicast address **224.0.0.6** for sending updates specifically to the Designated Router (DR) and Backup Designated Router (BDR).

13. Differentiate between classful and classless routing with reference to RIPv1.

RIPv1's limitations highlight the difference between these routing types.

- **Classful Routing (RIPv1 Feature):** Classful routing, used by RIPv1, means that periodic routing updates **do not carry subnet information**. This restricts the network, making it impossible to have different-sized subnets within the same network class (no support for VLSM).
- **Classless Routing (Contrast):** Classless routing (like CIDR) allocates addresses based on network prefixes and includes subnet mask information in updates, thereby **supporting Variable Length Subnet Masks (VLSM)** and allowing for efficient use of IP addresses across variable subnet sizes.